

UNIT IV — ALGAE

Definition — Algae: A broad traditional term for predominantly aquatic, photosynthetic, thalloid organisms. The syllabus treats them in the classical botanical framework.

4.1 General characteristics

- Mostly photosynthetic and predominantly aquatic.
- Body is commonly a thallus without true roots, stems and leaves.
- Organization ranges from unicellular to colonial, filamentous, siphonaceous and large parenchymatous forms.
- Reproduction can be vegetative, asexual and sexual.
- Pigments, reserve food and flagella help distinguish major groups in classical classification.

4.2 Thallus organization

Organization	Example type / idea
Unicellular motile	Single flagellated cell
Colonial	Cells organized into a colony, e.g., Volvox
Filamentous	Cells arranged in a chain
Siphonaceous/coenocytic	Large multinucleate tubular thallus, e.g., Vaucheria
Parenchymatous	More complex tissue-like organization in large seaweeds

4.3 Classical classification logic

In examination answers, use **pigments, reserve food, flagellar characters, cell-wall composition and thallus organization** as major criteria.

Major algal groups emphasized in the syllabus



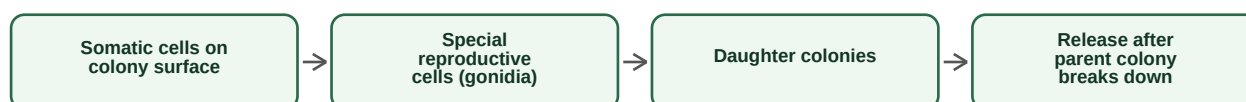
4.4 Chlorophyta — Volvox and Chara

Volvox

Definition — Volvox: A motile colonial green alga in which many biflagellate cells are embedded in a common gelatinous matrix and show functional specialization.

- Colony is generally spherical and hollow.
- Cells are connected and coordinated within the colony.
- Reproductive specialization may involve gonidia.
- Asexual daughter colonies and sexual reproduction are important study points.

Simplified Volvox asexual reproduction



Chara

Chara is a complex green alga with a plant-like appearance, differentiated main axis, nodes and internodes. Reproductive structures are highly distinctive and commonly studied morphologically.

4.5 Xanthophyta — Vaucheria

Definition — Vaucheria: A yellow-green alga with a siphonaceous, coenocytic tubular thallus.

- Multinucleate thallus with few cross walls in the vegetative body.
- Asexual reproduction may involve large zoospores in suitable species/conditions.
- Sexual reproduction is typically oogamous in the classical account.

4.6 Chrysophyta and Diatoms

- Golden-brown pigmentation is associated with characteristic accessory pigments.
- Diatoms possess a siliceous cell wall forming two overlapping halves called the frustule.
- Diatom forms and ornamentation are useful in identification.

Conceptual logic of the diatom frustule and size restoration



4.7 Phaeophyta — Ectocarpus

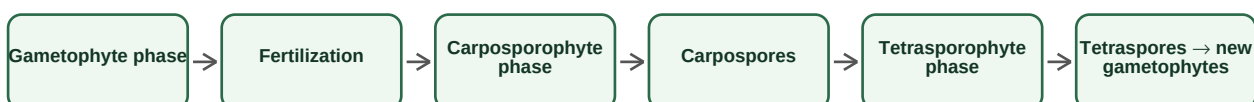
Ectocarpus is a representative brown alga used to study filamentous thallus organization and life-cycle patterns. Brown colour is associated with accessory pigments that mask chlorophyll. Learn the prescribed classroom life-cycle diagram.

4.8 Rhodophyta — Polysiphonia

Definition — Polysiphonia: A representative red alga with a characteristic polysiphonous filament organization.

- The thallus has a central axial cell surrounded by pericentral cells.
- Red coloration is associated with accessory phycobilin pigments.
- The classical life cycle is important because it illustrates a complex multi-phase sequence.

Simplified Polysiphonia life-cycle framework



4.9 Economic importance of algae

Area	Importance
Ecology	Primary production, oxygen release and aquatic food webs
Food/feed	Some algae are edible or used as feed supplements
Industry	Sources of hydrocolloids and other useful products
Agriculture	Some seaweed products are used in soil/plant management

Area	Importance
Research	Model organisms and bioactive compound studies

Exam focus: Unit IV high-yield diagrams: Volvox colony, Chara reproductive structures, Vaucheria thallus/reproduction, diatom frustule, Ectocarpus life-cycle concept and Polysiphonia triphasic concept.